

AI WeatherWise

AI-Augmented Backend Development Project

PHASE 1 – BRAINSTORMING AND IDEATION

Team No.	05
Team ID	SWTID-2026-9139
Project Title	AI WeatherWise
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1. Brainstorming

The team conducted a structured brainstorming session to identify a backend project that solves a real-world problem while demonstrating Artificial Intelligence integration, secure authentication, database management, and REST API development. Several ideas were discussed before selecting AI WeatherWise as the final project because it combined practical usefulness with the technologies covered in the AI-Augmented Backend Development program.

Project Ideas Considered

- AI WeatherWise – Intelligent Weather Forecast and Recommendation System
- AI FAQ Assistant
- AI FitTrack
- AI BlogNest API
- AI StudyBuddy

Discussion Outcome

The brainstorming session helped the team compare different ideas based on feasibility, backend complexity, scalability, and AI integration. AI WeatherWise was selected because it demonstrates Node.js, Express.js, MongoDB, JWT authentication, Weather API integration, and AI-powered recommendations within one modular backend application

2. Idea Generation and Prioritization

The shortlisted ideas were evaluated using measurable criteria to identify the strongest project for implementation.

Evaluation Criteria	AI WeatherWise
Real-world usefulness	High
Backend complexity	High
AI integration	Yes
MongoDB suitability	Excellent
REST API support	Complete
Scalability	High

Reasons for Selecting AI WeatherWise

- Provides practical weather assistance.
- Uses Node.js, Express.js, MongoDB, and Weather APIs.
- Implements JWT-based authentication.
- Supports favourite location management.
- Generates AI-powered weather recommendations.
- Follows a scalable modular backend architecture.

3. Define Problem Statement

Problem Statement

Users often switch between multiple weather applications to obtain forecasts for different locations. Existing services provide weather measurements but do not always offer personalized recommendations while securely managing user preferences and favourite locations.

Proposed Solution

AI WeatherWise is a RESTful backend service developed using Node.js, Express.js, MongoDB, and Mongoose. It authenticates users using JWT, stores favourite locations, retrieves current weather information from external APIs, and generates AI-powered weather recommendations through a modular backend architecture.

Objectives

- Provide secure user registration and login.
- Retrieve current weather information through APIs.
- Store and manage favourite locations.
- Generate personalized weather recommendations.
- Maintain centralized error handling and scalable architecture.

4. Expected Outcome and Phase Summary

The Brainstorming and Ideation phase established a clear project direction for AI WeatherWise. The team finalized a practical backend solution that combines authentication, database persistence, weather retrieval, and AI-driven recommendations, creating a strong foundation for Requirement Analysis, Project Design, Project Planning, Development, Testing, Documentation, and Demonstration.

Expected Deliverables

- Clearly defined project idea.
- Prioritized solution with documented justification.
- Well-defined problem statement and objectives.
- Strong foundation for the remaining seven project phases.